

The main goal that 5G aims to do as a technology can be divided into three groups:

- 1. Enhanced Mobile Broadband (eMBB):** Higher data-rates are specified. For the downlink, up to 100 Mbps are offered for outdoor and 1 Gbps for indoor (5GLAN), with half of these values available for the uplink. A number of case studies have been under consideration, amongst them is aviation – where eMBB is helping deliver a bitrate of 1-2 Gbps to an airborne flight.

KPI of eMBB in 5G networks are:

- peak data rates: 20 Gbps in Downlink (DL), 10 Gbps in Uplink (UL),
- user experienced data rates (5th percentile user throughput): 100 Mbps (DL), 50 Mbps (UL),
- area capacity: 10 Mbps/m² (indoor hotspot),
- user plane latency: 4 ms (average one-way).

eMBB Use Cases:

- To **stream high-definition video**, providing access to high-quality content without buffering or interruptions.
- Provides the necessary speed and capacity to maintain reliable Internet of Things (IoT) connections, allowing for the efficient collection and sharing of data.
- Enables immersive **Virtual Reality (VR) experiences**, providing smoother, lag-free experiences.
- High speeds and low latency enables competitive gaming experiences which allowing gamers to play against each other from anywhere.
- Allows for remote monitoring and telemedicine services, giving patients access to healthcare from any location.

- 2. Ultra Reliable and Low Latency Communications (uRLLC):** uRLLC is a key feature of 5G that enables high-reliability and low-latency communication for mission-critical applications such as factory automation, autonomous driving, and virtual/augmented reality. It provides a guaranteed quality of service (QoS) with extremely low latency (1 ms) and high reliability. For instance, for remote control of process automation, a reliability of 99,9999% is expected, with a user experienced data rate up to 100 Mbps and an end-to-end latency of 50 ms. This is provided in particular through the Edge Computing capability.

KPI of uRLLC in 5G networks are:

- user plane latency: less than 1 ms and reliability of 99.999% (one-way UL and DL),
- control plane latency: tens of ms,
- mobility interruption time: 0 ms.

uRLLC Use Cases:

- **Autonomous Vehicles:** uRLLC technology enables reliable data transmission and allows for real-time communication between vehicles.
- **Industrial Automation:** uRLLC provides reliable, low latency communication between machines and enables real-time control of processes.
- **Remote surgery:** uRLLC enables reliable data transmission for real-time communication between doctors and patients, allowing for more accurate diagnosis and treatment.
- **Augmented/Virtual Reality:** uRLLC ensures a smooth and immersive user experience with reliable data transmission for real-time communication
- **Smart Grids:** uRLLC allows for real-time monitoring and control of the grid, ensuring a stable and reliable power supply.

- 3. Massive Machine Type Communications (mMTC):** Several scenarios require the 5G system to support very high traffic densities of devices. It is about providing wireless connectivity to tens of billions of often low complexity, low-power machine-type devices with limited data rates.

KPI of mMTC in 5G networks are::

- connection density: 106 devices/km²,
- coverage: 164 dB Maximum Coupling Loss (MCL),
- device battery life: 10-15 years.

mMTC Use Cases:

mMTC will enable a variety of real-world use cases, from intelligent transportation systems to autonomous factories and smart energy grids. With its low-latency connections and high capacity, mMTC will be able to support large, densely populated areas with a massive number of devices. This will enable improved monitoring and control of public safety, the environment, and infrastructure, as well as enable faster, smarter, and more efficient cities.

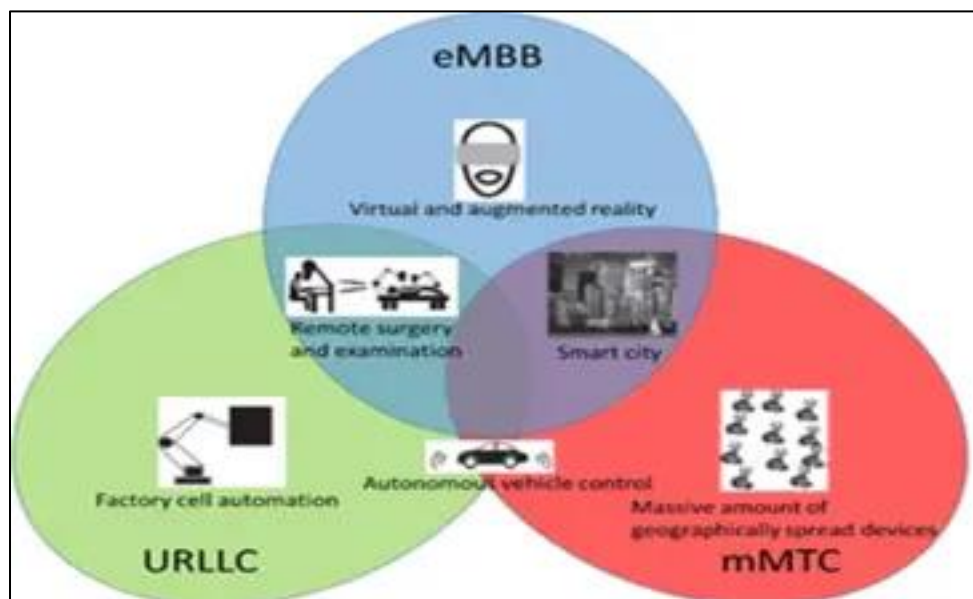


Figure: The main 5G use cases and applications